

Maxwell Newsome

Surrey, BC

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TECHNICAL SKILLS

Programming & Development: C, C++, Python, Java, SystemVerilog, Assembly, Bash, MATLAB

Tools: Git, Altium, LTspice, PSpice CAD, oscilloscopes, voltmeters, multimeters, Questa

Hardware: FPGA, STM32, Arduino, L298N motor drivers, IR sensors

Communication Protocols: I2C, CAN, SPI, TCP, UART

EDUCATION

University of British Columbia

Expected Graduation May 2028

Bachelor of Applied Science in Computer Engineering

TECHNICAL EXPERIENCE

AgroBot Design Team, University of British Columbia
Firmware Engineer

September 2025 - Present
Vancouver, BC

- Develop embedded firmware in C++ on an STM32 microcontroller to perform data acquisition of battery voltage and current using the internal ADC to support SOC and SOH estimation for a battery management system
- Designed an analog front-end (AFE) circuit integrating a voltage divider, Hall effect sensor, and low pass filter for signal conditioning prior to STM32 ADC sampling, using Altium for schematic capture and PCB layout
- Transmit calibrated telemetry data over UART and CAN protocols for integration with robot control and diagnostics

TECHNICAL PROJECTS

Multithreaded HTTP/1.1 Web Server, Personal Project

October 2025

- Built a multithreaded static file server in C++ over TCP using a thread pool with mutex-guarded task queue and condition variable synchronization for concurrent client handling
- Parsed HTTP requests to serve files from disk with proper headers and 404 handling, with a thread-safe LRU cache using a hash map paired with a doubly linked list to achieve O(1) get, insert, and eviction, reducing repeated disk I/O
- Implemented a thread-safe logging system tracking timestamps, status codes, response times, and thread IDs

Smart Soldering Air Quality Monitor, Personal Project

March 2026

- Designed and implemented an embedded air quality monitoring system on an STM32 Nucleo using FreeRTOS, integrating a BME680 sensor (I2C) and SPI-driven LCD for real-time environmental feedback
- Developed a multi-task architecture with mutex and semaphore synchronization to prevent race conditions across sensor acquisition, display updates, and high-priority alert handling tasks
- Created a threshold-based air quality classification with time-based escalation logic and a two-stage safety alert system using LED and PWM buzzer, with interrupt-driven user acknowledgment to reset persistent hazard states

Custom Dynamic Memory Allocator & Heap Visualizer, Personal Project

March 2026

- Implemented a custom dynamic memory allocator in C supporting alloc, free, and realloc with block splitting and bidirectional coalescing using header-footer metadata
- Built a heap simulation framework that executes allocation instructions from a script and logs block offsets, sizes, and allocation states for debugging and analysis
- Developed a Python script with Matplotlib to animate heap state changes over time, illustrating fragmentation and coalescing behavior for performance analysis of first, best and next-fit allocating algorithms on various heap sizes

Embedded Task Scheduler & VGA Game System, Personal Project

December 2025 - January 2026

- Programmed a bare-metal multitasking scheduler in C using machine-timer interrupts to coordinate UART, LED, VGA, and game tasks allowing tasks to run concurrently in a time-sliced manner
- Developed a memory-mapped VGA graphics engine with custom pixel, text, and shape rendering, including an interactive Snake.io-style game where the user competes against a programmed robot opponent
- Integrated hardware interrupts and switches to control task execution, game speed, and real-time system behavior, using a finite state machine with interrupt-driven input handling to start, stop, and switch tasks on an FPGA platform

WORK EXPERIENCE

Tong Louie YMCA

Lifeguard/Swim Instructor

August 2022 - Present
Surrey, BC

- Supervise high-traffic pool environments, ensuring 100% adherence to safety regulations
- Deliver swim instruction tailored to children's varying skill levels, improving confidence and technique

Haywood Securities

Compliance Intern

May 2025 - August 2025
Vancouver, BC

- Resolved discrepancies by liaising with investment advisors and blocking accounts when required